

Nicolás Garavito-Camargo – Curriculum Vitae

Flatiron Research Fellow | [Center for Computational Astrophysics \(CCA\)](#), NYC
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Research Interests: *Galactic Dynamics – Astrophysical probes of Dark Matter – Computational Methods – High-Performance Computing – N-body Simulations – Software development*

Education

Ph.D., Astronomy and Astrophysics, [University of Arizona](#), 2021.

Advisor: [Dr. Gurtina Besla](#)

M.Sc., Physics, [Universidad de Los Andes](#), Bogotá, Colombia, 2015.

Advisor: [Dr. Jaime E. Forero-Romero](#)

B.Sc., Physics, [Universidad Nacional de Colombia](#), Bogotá, Colombia, 2013.

Advisor: [Dr. Rigoberto A. Casas Miranda](#)

Appointment

Flatiron Research Fellow, Flatiron Institute, October 2021-December 2024.

Scholarships and Awards

- Barbara Pichardo Future Faculty Fellowship University of Arizona 2025-2027.
- University of Arizona, Theoretical Astrophysics Program, Graduate Student Research Prize 2021.
- University of Arizona College of Science award for Excellence in Service for graduate students, 2020.
(Awarded to one graduate student across the college of science per year.)
- University of Arizona theory travel grant, 2016, 2018.
- McCarthy-Stoeger Scholarship 2015-2017, Vatican Observatory.

Students supervision

I have had the privilege of advising or co-advising a total of eight students (five Ph.D., three undergraduates) from diverse backgrounds.

- **Richard Brooks** (Graduate Student at University of College London), Spring 2024 - present. I am advising Richard on one research project that was submitted for publication to ApJ (see article 29 in the publications section). We are currently working on a second paper, expected for submission in Spring 2025.
- **Elise Darragh-Ford** (Graduate student at Stanford), Fall 2023-Summer 2024. I advised Elise on a research project that was the final chapter of her Ph.D. thesis, resulting in one paper currently under preparation.
- **Silvio Varela** (Graduate student at Universidad de la Serena), Fall 2022-present. I am currently co-advising Silvio on a research project that will lead to a paper now in preparation
- **Arpit Arora** (Graduate student at University of Pennsylvania); Fall 2021-present. I am currently co-advising (now Dr.) Arpit on two research projects: one paper has been accepted to ApJ (see article 27th in the publications section), and another is in preparation.
- **Hayden Foote** (Graduate student at University of Arizona), Fall 2021-present I co-advised Hayden on a research project that resulted in a submitted publication (see article 28th in the publications section), and a second paper is currently in preparation.
- **Andrew Eden** (Undergraduate at Florida Institute of Technology), Fall 2022-Fall 2023. I advise Andrew on his undergraduate thesis project which will result in a paper that is currently in preparation. All the code that Andrew develop for his project is publicly available on [Github](#).

- **Ludia Adhikary** (Undergraduate CUNY), supervised through the AstroCOM CUNY/CCA program; NYC Summer 2022-Summer 2023. Ludia presented her work at the CCA symposium and she presented two posters. One at the SACNAS conference in 2022 and one at the winter AAS of 2023.
- **Stephanie Carolina Cely Rodriguez** (Undergraduate at Universidad Nacional de Colombia), I advise undergraduate Stephanie on her undergraduate thesis as part of the RECA summer program in Summer 2022. Stephanie successfully graduated in Summer 2023. A talk contribution from Stephanie was given at the RECA symposium and can be watch [here](#).

Teaching Experience

Principal lecturer of 3 courses taught for a total of 12 times. Teaching assistant for 3 courses.

- Guest Lecturer for the graduate Galaxies class, Columbia University, Fall 2022.
- Teaching assistant for the Astronomy Tutoring for Majors & Minors Program. University of Arizona, Spring 2019.
- Teaching assistant for the Computational Physics class PHYS305. University of Arizona. Spring 2018.
- 3 times Lecturer of the [Computational Tools](#) at Universidad de los Andes, spring semester 2014 - spring 2015.
- 3 times Lecturer of the computational Methods Laboratory at Universidad de los Andes, semester spring 2014 - spring 2015.
- Teaching assistant of the [Computational Methods](#) Universidad de los Andes, fall semester 2014.
- Lecturer (for 3 different sessions) of the class Physics I Laboratory at Universidad de los Andes, fall semester 2013.

Academic Service

- Lead coordinator of the EXP collaboration. I coordinate the cosmological simulations working group. I am in charge of 3 graduate students. Other responsibilities include: organizing in person meetings twice a year.
- Science organizing committee of the [Clouds over the Pyrenees](#) conference to be held in Benasque, Spain from September 6-9, 2026.
- Science organizing committee of the XMC workshop in Yellowstone to be held in May 26th-30th 2025, at the University of Montana.
- Local and Science organizing committee of the [Milky Clouds over Manhattan conference](#).
- PhD. thesis committee of Dr.Silvio Varela (University of La Serena, Chile), Nov-2023.
- Referee for: Astrophysical Journal (ApJ), Monthly Notices of the Royal Astronomical Society (MNRAS), Nature, Nature Astronomy, Astronomy and Astrophysics (A&A), and the Galaxies Journal.
- Beyond-BFE collaboration coordinator: I lead the cosmological simulation group, organize in person and online meetings.
- Mentor at the City University of New York CUNY-CCA program for undergraduate students at CUNY working with mentors at the CCA, summer 2022.
- Proposal reviewer for [Colombian Science Clubs](#), 2018.
- Local Organizing Committee of the [Andean Cosmology School](#), Universidad de Los Andes, 2015.
- Organizer of the student astronomy seminar at the Planetarium of Bogotá, 2014-2015.

Diversity, Equity, Inclusion (DEI) and outreach leadership:

DEI Leadership:

I have worked extensively to build supportive communities. I co-created a support network for astronomy students in Colombia, [RECA](#), which has grown to 478 members, making it the largest student association in astronomy in Latin America. In 2020, I established a yearly mentorship program designed to support students in their careers. To date, we have worked with 141 students, each paired with a senior mentor for guidance. Additionally, we have

compiled valuable resources, including career panels with guest speakers and online guides for applying to PhD programs. I have also secured funding to run this program through the IAU.

- Co-creator of the 10-week [RECA internship program](#) for astronomy students in Colombia. May-August 2021.
- Co-creator of the [RECA mentorship program](#) for astronomy students in Colombia. 2020-2024. (The mentorship program pairs up students with professional astronomers to provide guidance through the application process to graduate programs)
- Career panelist for CUNY undergraduates in stem “an initiative to enable low-income, talented domestic students to pursue successful careers in promising STEM fields” organized by Professor Viviana Acquaviva. December 3rd 2021.
- Co-organizer, [Diversity Journal Club](#) Steward Observatory, 2018-2021.
- Creator, [Astrocharlas](#), Steward Observatory, 2018-present. (Spanish outreach series talks in astronomy)
- Writer for [Astrobitos](#), 2018-present.
- Mentor, Tucson Initiative for Minority Engagement in Science and Technology Program [TIMESTEP](#), 2016-2018.

Outreach

- NYC Prison outreach program at MDC Brooklyn. Lead a 3-hour outreach session with Spanish speaking women. May 13, 2024 (another session will take place on November 6th). Program organizer: Kiyan Tavangar (U. Columbia)
- Classroom astronomer, NOAO Project [ASTRO](#), 2018-2019.
- Discussion leader, NOAO [Teen Astronomy Cafe](#), 2018.
- Planetarium SpaceArt mentor for Children, Bogotá, Colombia, 2013-2014.

Open source and HPC experience

- Core developer and active contributor of the python packages: (Core developer) [EXPtools](#), (Core developer) [NBA](#), Cranes. (Contributor) [py-Ananke](#), (Contributor) [gala](#), (Contributor) [halo analysis](#), (Contributor) [gizmo analysis](#).
- Expertise with HPC N-body codes: Gadget-3, 4, EXP (CPU & GPU), AREPO.

Scientific Talks

47 Total: 18 Invited (denoted by †), 26 in North America, 6 in Europe, 9 in Latin America, 1 Asia.

Conferences (14)

- KITP workshop “Dark Matter Theory, Simulation, and Analysis in the Era of Large Surveys”, UC Santa Barbara, June 2024. †
- The Milky Way is not an island, Sexten, February, 2024.
- Surveying the Milky Way: The Universe at our backyard. Caltech, Pasadena, October, 2023.
- Friends of Friends meeting, Cordoba, Argentina, April, 2023. †
- IAU 379: Dynamical masses of local group galaxies, contributed talk, March, 2023.
- [IAU 377: Early Disk-Galaxy Formation from JWST to the Milky Way](#), contributed talk, February, 2023.
- [Colombian Congress of Astronomy](#), Plenary talk, August, 2022. †
- [Friends of friends meeting](#), Cordoba, Argentina, April, 2022.
- [Division on Dynamics Astronomy](#), Virtual meeting, May 2021.
- [Streams 21](#), Virtual Conference, February 2021.

- [The Local Group: Assembly and Evolution](#), virtual conference, August 2020.
- [European Astronomical Society meeting](#), virtual meeting, June 2020.
- Durham University, UK, [Small Galaxies Cosmic Questions](#), August, 2019.
- MPA, Heidelberg, [Stellar halos across the cosmos](#), July 2018.
- LARIM, Cartagena, Colombia, October 2016.
- EWASS, Geneva, Switzerland, July 2014.

Seminars and Colloquia (33)

- UC Santa Barbara, seminar, June 2024. †
- KITP UC Santa Barbara, seminar, June 2024. †
- UC Riverside, seminar, October 2023. †
- Universidad de La Serena, La Serena, Chile, May 2023. †
- Instituto de Astronomía y Física del Espacio (IAFE), Buenos Aires, April, 2023.
- Max Planck Institute for Astrophysics, Cosmology Seminar, March, 2023.
- U. Rutgers, Astronomy Seminar, Nov, 2022. †
- U. Columbia, Lunch talk, Sept, 2022. †
- STScI, galaxies lunch talk, May, 2022. †
- Universidad de Antioquia, seminar, February, 2022. †
- University of Massachusetts, Amherst, Colloquium, Jan, 2022. †
- University of Madison-Wisconsin, Science seminar, November 2021.†
- University of Michigan, Galaxies group seminar, November 2021.†
- NYU, CCPP, seminar, November 2021.†
- CCA, Flatiron Institute, Lunch Talk, October, 2021.
- Steward Observatory, Theoretical Astrophysics Program (TAP) colloquium, September, 2021.†
- [ComSciCon](#), June 2021. †.
- Steward Observatory, Galaxy lunch talk, March, 2021.
- Univesidad de los Andes, Astronomy Seminar, February, 2021.
- UC Irvine, Astronomy Seminar, January, 2021. †
- Steward Observatory, Early Career Scientist talk, December, 2020. †
- CCAPP, Seminar, December, 2020. †
- Princeton, Journal Club, November 2020. †
- KIPAC, Stanford, Tea-Talk, October 2020.
- Harvard Center for Astrophysics, GCSP seminar, October 2020.
- Carnegie Observatories, seminar, October 2020. †
- University of California Berkeley, lunch talk, September 2020.
- The Royal Observatory of Edinburgh, UK, Coffee Talk, August 2019. †
- W.M. Keck Observatory, Journal Club, June 2019.
- Magellanic Cloud Fest III, University of Arizona, May, 2019.
- JILA Seminar, JILA Institute, University of Colorado, December 2017.
- STScI Galaxies Journal Club. December 2016.
- Magellanic Cloud Fest II, University of Arizona, March 2016.
- UNAM, Mexico City, Mexico, Nov 2013.
- Centro de Investigaciones De Astronomía CIDA, Merida, Venezuela, July 2013.

†Invited

Posters

- European Astronomical Society meeting, virtual meeting, June 2020.
- Rediscovering our Galaxy, IAU symposium 334, Potsdam, Germany. July 2017.

Telescope and HPC time Awarded

- Hubble Space Telescope, 32 orbits, Cycle 31, 2023. PI: Prof. Sukanya Chakrabarti (U. Alabama).
- MareNostrum Super computer, 1.7 million CPU hours, AECT-2023-2-0016, 2023. PI: Prof. Chervin Laporte (U Barcelona).
- GMRT 60 hours cycle 44, 2023. PI: Prof. Karin Menendez-Delmestre (Valongo Observatory, Rio de Janeiro).
- Blanco Telescope, “*A VISTA-DECam Experiment in Near-Field Cosmology: Search for the Magellanic Dark Matter Wake*” cycle 2020B. PI: Prof. Julio Chaname (Universidad Catolica de Chile), 3 nights.

Grants

- LSSTC Grant Award (Virtual Internship in Rubin/LSST Science to Provide Research Experience to Undergraduate Students in Colombian Institutions) **2021-51**, 2021. CO-PI: Nicolás Garavito-Camargo. This grant was awarded to support undergraduate research summer projects.
- International Astronomical Union (IAU) Office of Astronomy development (OAD) **grant 2021**. PI: Nicolás Garavito-Camargo. This grant was awarded to support the **mentorship program for undergraduate students** in Colombia.
- University of Arizona theory travel grant, 2016 and 2018.
- IAU travel grant, 2016.

Observing Experience

- DECam, Blanco-4m telescope, CTIO, Chile, 3 nights, 2020.
- VATT telescope. Mt Graham, Arizona, 4 nights, 2016.
- CIDA, Merida, Venezuela - 2 nights, 2013.

Research highlights in the news

- Sky & Telescope: How our largest dwarf galaxy keeps other in line
- JPL Nasa: Astronomers Release New All-Sky Map of Milky Way’s Outer Reaches
- Syfy Wire: Dark Matter could be powering a galaxy that orbits the Milky Way until they collide
- Astrobites:
- Phys.org: Astronomers release new all-sky map of the Milky Way’s outer reaches
- University of Arizona news: Astrophysicist help chart dark matters invisible ocean
- AAS NOVA 2020: An Asymmetric Dark Matter Halo
- AAS NOVA 2019: Hunting for a Dark Matter Wake

Publications list

Refereed: 30 – First author: 5 – Supervised students: 2 (denoted by †) – Co-supervised students (denoted by ‡)
h-index: 14 – citations : 1121 (as of December 1st, 2024)

[ORCID](#), [ADS](#), [arXiv](#), [Google Scholar](#)

First author publications or with significant contributions as mentor

30. *Implications for a High Mass M31: M33's Orbital History and M31's Response to the Passage of M33.* Ekta Patel, **Nicolás Garavito-Camargo**, Ivanna Escala. ApJ Submitted 2024. (paper available upon request)
29. *†LMC Calls, Milky Way Halo Answers: Disentangling the Effects of the MW-LMC Interaction on Stellar Stream Populations* Richard A. N. Brooks, **Nicolás Garavito-Camargo**, Kathryn V. Johnston, Adrian M. Price-Whelan, Jason L. Sanders, Sophia Lilleengen. ApJ in press 2024. [2 citations]
28. *‡Segue 2 Recently Collided with the Cetus-Palca Stream: New Opportunities to Constrain Dark Matter in an Ultra-Faint Dwarf* Hadyn R. Foote, Gurtina Besla, **Nicolás Garavito-Camargo**, Ekta Patel, Guillaume F. Thomas, Ana Bonaca, Adrian M. Price-Whelan, Annika H. G. Peter, Dennis Zaritsky, Charlie Conroy. ApJ in press 2024.
27. *†LMC-driven anisotropic boosts in stream-subhalo interactions.* Arora, A., **Garavito-Camargo, N.**, Sanderson, R. E., Cunningham, E. C., Wetzel, A., Panithanpaisal, N., Barry, M. ApJ 974, 2, 2024. [8 citations]
26. *On the co-rotation of Milky Way satellites: LMC-mass satellites induce apparent motions in outer halo tracers* **Nicolás Garavito-Camargo**, Adrian M. Price-Whelan, Emily C. Cunningham, Jenna Samuel, Ekta Patel, Andrew Wetzel, Kathryn V. Johnston, Arpit Arora, Robyn E. Sanderson, Lehman Garrison, and Danny Horta. ApJ, 975, 1, 2023. [8 citations]
25. *The Clustering of Orbital Poles Induced by the LMC: Hints for the Origin of Planes of Satellites* **Garavito-Camargo, Nicolás**; Patel, Ekta; Besla, Gurtina; Price-Whelan, Adrian; Laporte, Chervin; Gómez, Facundo A; Kathryn V. Johnston; ApJ 923, 2, 2021. [30 citations]
24. *Quantifying the impact of the Large Magellanic Cloud on the structure of the Milky Way's dark matter halo using Basis Function Expansions* **Garavito-Camargo, Nicolás**; Besla, Gurtina; Laporte, Chervin; Price-Whelan, Adrian M.; et al. ApJ, 919, 109, (2021). [95 citations]
23. *Quantifying the Stellar Halo's Response to the LMC's Infall with Spherical Harmonics.* Cunningham, Emily C; **Garavito-Camargo, Nicolas**, Deason, Alis J; Johnston, Kathryn V. et al. ApJ, 898, 1,(2020). [52 citations]
22. *Hunting for the DM Wake induced by the LMC.* **Garavito-Camargo, Nicolas**; Besla, Gurtina; Laporte, Chervin F.P; Johnston, Kathryn V; Gómez, Facundo A; Watkins, Laura. ApJ Accepted, (2019). [162 citations]
21. *The impact of gas bulk rotation on the morphology of the Lyman-alpha line.* **Garavito-Camargo, J.N**; Forero-Romero J.E; Dijkstra M. ApJ, 795, 120, (2014). [10 citations]

Publications with significant contributions

20. *The distant Milky Way halo from the Southern hemisphere: Characterization of the LMC-induced dynamical-friction wake.* Manuel Cavieres, Julio Chanamé, Camila Navarrete, Yasna Ordenes-Briceño, **Nicolás Garavito-Camargo**, Gurtina Besla, Maren Hempel, Katherina Vivas, Facundo Gómez (ApJ in press 2024).
19. *The All-Sky Impact of the LMC on the Milky Way Circumgalactic Medium* Christopher Carr, Gerg L. Bryan, **Nicolás Garavito**, Gurtina Besla, David J. Setton, Kathryn V. Johnston (ApJ submitted 2024).
18. *All-Sky Kinematics of the Distant Halo: The Reflex Response to the LMC* Vedant Chandra, Rohan P. Naidu, Charlie Conroy, **Nicolas Garavito-Camargo**, Chervin F. P. Laporte, Ana Bonaca et al. (Submitted to ApJ 2024).
17. *Structure, Kinematics, and Observability of the Large Magellanic Cloud's Dynamical Friction Wake in Cold vs. Fuzzy Dark Matter* Hayden R. Foote, Gurtina Besla, Philip Mocz, **Nicolás Garavito-Camargo**, Lachlan Lancaster, Martin Sparre, Emily C. Cunningham, Mark Vogelsberger, Facundo A. Gómez, and Chervin F. P. Laporte, ApJ submitted 2023

16. *Implications of the Milky Way travel velocity for dynamical mass estimates of the Local Group*
Katie Chamberlain, Adrian M. Price-Whelan, Gurtina Besla, Emily C. Cunningham, **Nicolás Garavito-Camargo**, Jorge Peñarrubia, Michael S. Petersen. ApJ (2022)
15. *Detection of the All-Sky Response of the Galactic Halo to the Magellanic Clouds*
Conroy, Charlie; Naidu, Rohan P; **Garavito-Camargo; Nicolás**; Besla, Gurtina; et al. Nature (2021).
14. *The orbital histories of Magellanic Satellites Using Gaia DR2 proper motions.*
Patel, Ekta; Kallivayalil, Nitya; **Garavito-Camargo, Nicolas** et. al., ApJ 893, 121, (2020).
13. *Modelling the gas kinematics of an atypical Ly α emitting compact dwarf galaxy.*
Forero-Romero, Jaime E., Gronke, Max., Remolina-Gutiérrez, Maria Camila, **Garavito-Camargo, Nicolas**, Dijkstra M. MNRAS, 474, 12F, (2018).

Publications with moderate contributions

12. *Efficient and accurate force replay in cosmological-baryonic simulations* Arpit Arora, Robyn Sanderson, Christopher Regan, **Nicolás Garavito-Camargo**, Emily Bregou, Nondh Panithanpaisal, Andrew Wetzel, Emily Cunningham, Sarah Loebman, Adriana Dropulic, Nora Shipp (Submitted to ApJ.2024)
11. *Generating synthetic star catalogs from simulated data for next-gen observatories with py-ananke*
Adrien C. R. Thob, Robyn E. Sanderson, Andrew P. Eden, Farnik Nikakhtar, Nondh Panithanpaisal, **Nicolás Garavito-Camargo**, and Sanjib Sharma (In press JOSS 2024).
10. *Dark matter distribution in Milky Way-analog galaxies*
Natanael Gomes-Oliveira, K. Menéndez-Delmestre, T. S. Gonçalves, D. C. Rodrigues, M. Grossi, **N. Garavito-Camargo**, A. Araújo, P. P. B. Beaklini, Y. Cavalcante-Coelho, A. Cortesi, L. H. Quiroga-Núñez, T. Randriamampandry (ApJ 2023).
9. *The proto-galaxy of Milky Way-mass haloes in the FIRE simulation*
Horta, Danny; Cunningham, Emily C.; Sanderson, Robyn; Johnston, Kathryn V.; Deason, Alis; Wetzel, Andrew; McCluskey, Fiona; **Garavito-Camargo, Nicolás**; Necib, Lina; Faucher-Giguère, Claude-André ; Arora, Arpit ; Gandhi, Pratik J. (ApJ Submitted 2023)
8. *Galactoseismology in cosmological simulations: Vertical perturbations by dark matter, satellite galaxies and gas.* Garcia-Conde. B, Antoja. T, Roca-Fabrega. S, Gómez. G, Ramos. P., **Garavito-Camargo. N**, Gómez-Flechoso, MA. (accepted for publication in MNRAS 2023)
7. *The impact of the Large Magellanic Cloud on dark matter direct detection signals*
Smith-Orlik, Adam ; Ronaghi, Nima ; Bozorgnia, Nassim ; Cautun, Marius ; Fattahi, Azadeh ; Besla, Gurtina ; Frenk, Carlos S. ; **Garavito-Camargo, Nicolás** ; Gómez, Facundo A. ; Grand, Robert J. J. ; Marinacci, Federico ; Peter, Annika H. G. JCAP submitted 2023
6. *Lopsided Galaxies in a cosmological context: a new galaxy-halo connection*
Silvio Varela-Lavin, Facundo A. Gómez, Patricia B. Tissera, Gurtina Besla, **Nicolás, Garavito-Camargo**, Federico Marinacci. Submitted to MNRAS 2022.
5. *On the stability of tidal streams in action space*
Arpit Arora, Robyn E. Sanderson, Nondh Panithanpaisal, Emily C. Cunningham, Andrew Wetzel, **Nicolás Garavito-Camargo**. ApJ l, vol. 939, no. 1, (2022).
4. *The highest-speed local dark matter particles come from the Large Magellanic Cloud.*
Besla, Gurtina; Peter, Annika; **Garavito-Camargo, Nicolas**. JCAP, 11, 13, (2019).
3. *The influence of Sagittarius and the Large Magellanic Cloud on the stellar disc of the Milky Way Galaxy.*
Laporte, Chervin F. P; Johnston, Kathryn V; Gómez, Facundo A; **Garavito-Camargo, Nicolas**; Besla, Gurtina. MNRAS, 481, 286L, (2018).

2. *The Extremely Luminous Quasar Survey in the Sloan Digital Sky Survey Footprint. II. The North Galactic Cap Sample.*
Schindler, Jan-Torge; Fan, Xiaohui; McGreer, Ian D; Yang, Jinyi; Wang, Feige; Green, Richard; **Garavito-Camargo, Nicolas** et al., ApJ, 863, 144S, (2018).
1. *Response of the Milky Way's disc to the Large Magellanic Cloud in a first infall scenario.*
Laporte, C.; Gomez, F; Besla, Gurtina; Johnston, Kathryn V; & **Garavito-Camargo, Nicolas**. MNRAS, 473, 1218L, (2018).

White papers

3. *NANCY: Next-generation All-sky Near-infrared Community survey* Y Jiwon Jesse Han, et al. (incl. **Garavito-Camargo, N**). [call for white papers for the Roman Core Community Survey](#)
2. *Mass Spectroscopy of the Milky Way*
Dey, Arjun. et al., (incl. **Garavito-Camargo, N**). [Astro2020: Decadal Survey on Astronomy and Astrophysics, Vol. 51, Issue 3, id. 489 \(2019\).](#)
1. *The Multidimensional Milky Way.*
Sanderson, Robyn E .et al. (incl. **Garavito-Camargo**) 2019. [Astro2020: Decadal Survey on Astronomy and Astrophysics, 2019, Vol. 51, Issue 3, id. 347 \(2019\).](#)